

Business Architecture

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Business Architecture

Introduction

The business architecture describes how a statistical organisation is structured to turn data into official statistics: which business domains exist, which business functions and processes they perform, which business objects they create and exchange, and who is accountable for them. Where the Strategy layer answers what the organisation must be able to do (capabilities) and what value it creates (value streams), the business architecture answers how that work is organised and where responsibility for data passes from one part of the organisation to another.

That last question is central for official statistics. A statistical value chain is long, involves many teams, and processes data that is often confidential. Errors, untraceable changes, or uncontrolled reuse at any point undermine trust in every statistic built on top of it. The SAF business architecture therefore pays explicit attention to the points at which data is persisted, quality-assured, and handed over. These handover points form the backbone of every reference model in this chapter.

The Statistical Proces

The statistical process is divided into three main business domains (observation, processing/analysis, and dissemination), with processing/analysis further divided into four subdomains. This results in a total of six business domains:

- Observation/Collection
- Processing/analysis
 - Standardization without loss of content
 - Editing, derivation, and non-response correction
 - Estimation and closed-loop
 - Statistically composed and composite statistical products
- Dissemination

Each of the business domains use the following processes :

1. Design of the execution processes; provides metadata
2. Implementation of the execution processes; provides the execution process
3. Execution of the execution processes; provides the statistical data
4. Management of the execution processes; provides quality reports

Each domain focuses on adding a specific statistical value to some data. Once sufficient value has been added, a so-called quality version is versioned that must be kept and potentially shared to some consumers. This careful approach is crucial due to the growing interest in our microdata, combined with the increasing focus on privacy. If we combine the added values of the six business domains, a value chain of statistical data is created. To realize statistical value within a business domain, execution processes (including governance) have been designed and implemented. How an execution process adds statistical value, and is therefore designed and implemented, falls outside the scope of the B&I architecture. This framework must stem from, among other things, the methodology set.

Key concepts

Concept	Definition
Business domain	A coherent part of the statistical process that adds one specific kind of statistical value to data and is governed as a unit (e.g., Standardization).
Business function	A stable grouping of behaviour within a domain, independent of organisation and technology (e.g., <i>Classify Data</i>).
Data state	A defined condition of data, characterised by the processing it has undergone and the quality criteria it meets.
Steady state	A data state that is persisted, versioned, quality-assured, and registered so it can be handed over and reused.
Quality gate	The set of entry criteria data must meet before it may be registered in a state.
Quality version	An immutable, identified version of a dataset that has passed a quality gate.
Handover interface	The contract between a producing and a consuming domain: the artefact, its required metadata, the triggering event, and the transfer of responsibility.
Reference model	A generic arrangement of domains, states, and handover points that an NSI can adopt and adapt.

Reference Models for the Statistical Data Value Chain

Why the SAF offers more than one model NSIs differ in size, mandate, legal context, data volumes, and technology platform. A single prescribed arrangement would fit some offices well and others poorly. The SAF therefore describes reference models: alternative, internally consistent ways of organising the value chain. Each model answers the same four questions:

1. At which points is data persisted and made available for reuse?
2. Which quality criteria must data meet at each point?
3. Who is accountable for the data at each point?
4. How are confidentiality and traceability guaranteed along the chain?

The SAF currently describes two reference models. Both follow the same underlying idea, progressive improvement of data through defined layers, but they differ in granularity and in where they place governance.

Model A: Steady-State Architecture

The Steady-State Architecture organises the value chain into business domains that are loosely coupled through five numbered steady states (#1 Raw, #2 Standardized, #3 Processed, #4 Statistics, #5 Released). Each state represents a specific kind of statistical value and has its own quality gate, ownership, and handover interface. The model is derived from production practice in NSIs with many statistical domains and extensive microdata reuse. It is described in full in the remainder of this chapter.

Model B: Medallion Architecture

The Medallion Architecture organises data into three layers of increasing quality, usually called Bronze (raw), Silver (cleansed and conformed), and Gold (curated for consumption). The pattern originates from lakehouse platforms and is a logical design pattern rather than a product. Its strength lays in its simplicity. There are three layers, a clear direction of flow, and the ability to rebuild any layer from the one below it.

As a data-engineering pattern, Medallion does not by itself define statistical semantics, accountability, or confidentiality rules per layer.

Comparison

Characteristic	Steady-State Architecture	Medallion Architecture
Number of governed handover points	Five (#1 to #5)	Three (Bronze, Silver, Gold)
Origin	Statistical production practice	Lakehouse data engineering
Granularity of quality gates	Fine: one gate per kind of statistical value	Coarse: several kinds of value combined per layer
Separation of standardization and editing	Explicit (#2 vs. #3)	Combined in Silver
Separation of internal statistics and released data	Explicit (#4 vs. #5)	Combined in Gold unless subdivided
Governance overhead	Higher: more states to own, register, and audit	Lower
Fit with platform tooling	Platform-neutral; mapping required	Native in lakehouse platforms
Typical adopters	Larger NSIs, many domains, high microdata reuse	Smaller NSIs, platform-led modernisation

Selection criteria

The criteria below help an NSI decide which model fits best. They are indicative. No single criterion is decisive, and the SAF toolkit will provide a structured assessment.

Criterion	Points towards Steady-State	Points towards Medallion
Organisational size and number of statistical domains	Many domains, many teams handing over data	Few domains, small central data team
Reuse of intermediate microdata across domains	High; standardized and processed microdata reused by several products	Limited; most data flows within one product chain
Legal requirements on pseudonymisation timing	Law or policy requires a separate, auditable pseudonymisation point	Pseudonymisation can be applied at the Silver boundary
Archival and reproducibility obligations	Specific states (e.g., processed microdata) must be retained long term	Retention can be defined per layer
Separation of internal statistics from released data	Required (embargo, different confidentiality thresholds per use)	Achievable by subdividing Gold
Metadata and governance maturity	Mature data ownership and catalogue practices in place	Governance capacity is limited and must stay lean
Technology platform	Heterogeneous landscape, on-premises and private cloud	Single lakehouse platform
Participation in international data exchange	Many handover points to external partners	Exchange concentrated on final outputs

Practice within the SAF community indicates that smaller NSIs often adopt a Medallion arrangement, while larger NSIs with many domains tend towards finer-grained states. This observation is to be substantiated through the SAF toolkit assessments.

Combining the models

The models are not mutually exclusive. The steady states can be mapped onto Medallion layers, which allows an NSI to use Medallion as its platform convention while governing data through steady states, or to start with Medallion and introduce finer states as it grows.

Steady state	Medallion layer	Note
#1 Raw Data	Bronze	Direct correspondence
#2 Standardized Data	Silver (lower sub-layer)	In Medallion often merged with #3
#3 Processed Data	Silver	Main reusable microdata layer
#4 Statistics	Gold (internal sub-layer)	Not yet released
#5 Released Data	Gold (published sub-layer)	Released after final disclosure control

Steady-State Architecture

Overview

Business domains

The statistical process is divided into three main business domains (Observation, Processing/Analysis, and Dissemination). Processing/Analysis is divided into four subdomains, which results in six business domains. Each domain adds one specific kind of statistical value and produces one steady state.

#	Business domain	Value added	Business functions (ArchiMate view)	Value stream	GSBPM	State in → out
1	Observation/Collection	Data in NSI custody, authentic and traceable	Receive Data, Extract Data, Collect Survey	VS-01	4	#0 → #1
2	Standardization without loss of content	Common structure, format, and classifications	Transform Data, Anonymize Data, Classify Data	VS-02	5.1–5.2	#1 → #2
3	Editing, derivation, and non-response correction	Reliable, complete, enriched microdata	Imputate Data, Integrate Data	VS-03	5.3–5.5, 5.8	#2 → #3
4	Estimation and balancing	Representative statistics with accuracy	Estimate Data, Aggregate Data	VS-04	5.6–5.7, 6.1–6.4	#3 → #4
5	Statistically composed and composite products	Coherent, approved products	Compose Data	VS-05	6.5, 7.1–7.2	#4 → #4 (approved)
6	Dissemination	Disclosure-safe, accessible released data	Data Delivery, Data Publication	VS-06	6.4, 7.3–7.5	#4 → #5

#	Business domain	Value added	Business functions (ArchiMate view)	Value stream	GSBPM	State in → out
—	Consumption (external)	Use of statistics by society	Data Consumption, Consumption Optimization	VS-10 (feedback)	8	#5 → (#6)

The execution processes between the business domains are loosely coupled. Each has its own design and implementation. For the mutual exchange of data, they are loosely linked via steady state interfaces. To this end, five steady state phases are recognized as handover points at business-domain boundaries: 1. raw data, 2. standardized data without loss of content, 3. processed data, 4. statistics, and 5. released data.

A steady state phase is therefore an environment where high-quality versions of data are brought together for exchange, but it also represents a value in the value chain of this data.

The four process types

Each business domain uses the same four process types:

1. **Design** of the execution processes, which provides metadata.
2. **Implementation** of the execution processes, which provides the execution process.
3. **Execution** of the execution processes, which provides the statistical data.
4. **Management** of the execution processes, which provides quality reports.

How an execution process adds statistical value, and therefore how it is designed and implemented, falls outside the scope of the SAF business architecture. That design follows from, among other things, the NSI's methodology.

Design rules of the steady-state model

The model rests on six design rules. They apply the SAF architecture principles to the data value chain.

1. **Loose coupling through states (SAF-P07).** Domains exchange data only through registered steady states, never through each other's internal working data. A domain can change its methods or systems without affecting its neighbours, as long as the state it produces meets its quality gate.
2. **One state, one kind of value.** Each state represents exactly one kind of added statistical value. This keeps quality gates testable and accountability unambiguous.
3. **Immutable quality versions.** Data registered in a state is never overwritten. Corrections and redeliveries create a new version, so every statistic can be traced to the exact data it was built on.
4. **No state without metadata (SAF-P09).** Registration requires the minimum metadata set of the state. Data without it does not enter the state.
5. **Progressive confidentiality (SAF-P11).** Protection increases along the chain: direct identifiers are removed or pseudonymised at #2, disclosure risk is assessed at #3, initial disclosure control is applied at #4, and final disclosure control gates #5.
6. **Reuse before reprocessing (SAF-P08).** An authorised consumer that needs data at a given level of value reuses the registered state rather than repeating the upstream processing.

Steady states at a glance

Position	Name	Produced by	Kind of value	Mandatory?
(#0)	Source Data	External provider	Outside NSI custody, boundary position	n/a
#1	Raw Data	VS-01	Authentic, traceable custody	When the product collects its own data
#2	Standardized Data	VS-02	Comparability, values unchanged	Optional
#3	Processed Data	VS-03	Reliability and completeness of microdata	Always
#4	Statistics	VS-04, VS-05	Representative, approved statistics	Always
#5	Released Data	VS-06	Disclosure-safe, accessible output	When output is released
(#6)	Consumed Data	External users	Outside NSI control, boundary position	n/a

State lifecycle

Every steady state follows the same lifecycle:

1. **Promotion.** The producing domain submits a dataset to the quality gate of the target state.
2. **Gate outcome.** The dataset is accepted, conditionally accepted with explicit quality flags, or rejected. Rejected data does not enter the state; it returns to the producing domain or, for #1, to the provider.
3. **Registration.** An accepted dataset receives a persistent identifier, its metadata is registered in the catalogue, and a registration event notifies consuming domains.
4. **Consumption.** Authorised consuming domains read the registered version. Responsibility for further processing transfers on receipt of the event; custody of the registered version stays with the producer.
5. **Revision.** A correction produces a new version that passes the same gate. Consumers are notified and decide whether to reprocess. Earlier versions remain available for as long as statistics depend on them.
6. **Retirement.** A version is disposed of according to the retention policy of the state, after its lineage obligations have been met.

How each state is described

Each state below is described with the same ten attributes. NSIs can use this template to document their own implementation; the SAF toolkit will provide it as a fill-in form.

#	Attribute	Question it answers
1	Definition and value added	What is this state and why does it exist?
2	Producing and consuming domains	Who creates it, and who may use it?
3	Quality gate	What must data satisfy to be registered?
4	Data and metadata characteristics	What does the data look like, and which metadata must accompany it?
5	Security, privacy, and access	How is the data protected, and who may see it?
6	Ownership and governance	Who is accountable?

#	Attribute	Question it answers
7	Versioning, retention, and lineage	How are versions managed, how long are they kept, and how is the trace preserved?
8	Reuse and handover	How is the state made available to consumers?
9	Traceability to the framework	Which capabilities, value-stream stages, and GSBPM sub-processes relate to it?
10	Examples and anti-patterns	What does good practice look like, and what should be avoided?

Boundary Position #0: Source Data

Description: Source data is data as it exists at the data holder, before it enters the NSI: a register at a government agency, a company's accounting system, a respondent's answers before submission, or a web page. The NSI has no custody of source data and cannot guarantee its quality. It can influence it only through agreements, questionnaire design, and cooperation with the data holder.

Why it is described: The SAF describes #0 so that the boundary of the NSI's responsibility is explicit. Everything the NSI knows about source data is captured as metadata when the data crosses into #1: who provided it, under which agreement, through which channel, and in what condition.

State #1: Raw Data

1. Definition and value added

Raw data is data received from an external data holder or respondent, placed under NSI custody in its original form, and verified for technical integrity and authenticity. It is the first point of possession. At this moment the NSI does not have access to the data at the source, so #1 is the earliest point at which the NSI can guarantee what it holds.

The value of #1 is **authentic custody**. It proves that the NSI holds exactly what the provider delivered, when it was delivered, under which legal basis, and through which channel. That proof is the foundation for all downstream traceability. If a statistic is ever questioned, the chain of evidence ends here.

2. Producing and consuming domains

Role	Domain	Note
Producer	Observation/Collection (VS-01)	Functions: Receive Data, Extract Data, Collect Survey
Primary consumer	Standardization (VS-02)	Triggered by the <i>Raw Data Registered</i> event
Secondary consumer	Editing (VS-03)	Only where #2 is skipped for this product
Occasional consumer	Innovation and Methodology (VS-08)	Source exploration, under specific authorisation
Assurance	Governance and Compliance (VS-07)	Audit of provenance and legal basis

3. Quality gate

The gate for #1 checks technical and legal conditions only. It does not assess content; content quality is the concern of #3.

Criterion	Check	Typical outcome on failure
Legal basis	A valid agreement or statutory basis covers this delivery	Reject; escalate to legal
Sender authenticity	Sender verified (certificate, credentials, signed delivery)	Reject
Integrity	Checksum or digital signature matches	Reject; request redelivery
Security	Malware and security scan passed	Quarantine; reject
Structural conformance	Delivery matches the agreed technical specification (file format, schema, encoding)	Reject or conditionally accept with flag
Completeness against agreement	Record counts and delivery scope within agreed tolerance	Conditionally accept with flag, or reject
Timeliness	Delivered within the agreed window	Accept with flag; report to relationship management
Metadata	Minimum metadata set captured (see attribute 4)	Hold until complete

Conditional acceptance is a deliberate option. A late or slightly incomplete delivery may still be the best available data. The flag ensures every downstream user knows this.

4. Data and metadata characteristics

Data. Raw data is stored in its original format, or in a lossless technical equivalent. Permitted technical operations are those that change representation without changing content and that can be reversed: decryption, decompression, character-encoding normalisation, and container conversion. Operations that change values, codes, or record selection are not permitted at #1. The original byte stream, or a verifiable hash of it, is retained.

Minimum metadata set.

Metadata element	Description
Persistent identifier and version	Unique identifier of this delivery version
Provider	Organisation or respondent group that supplied the data
Agreement reference and legal basis	Agreement identifier and the applicable legal ground
Channel and method	Transfer channel (file transfer, API, data space connector, survey mode) and instrument version
Receipt timestamp	Moment of receipt into NSI custody
Integrity proof	Checksum or signature value and algorithm
Volume	Record count and size
Reference period	Period the data describes, as stated by the provider
Structural description	Schema or technical specification the delivery conforms to
Gate outcome	Accepted or conditionally accepted, with quality flags
Classification	Availability, integrity, and confidentiality rating

In GSIM terms, #1 typically instantiates a *Data Set* received from an *Information Provider* through an *Exchange Channel* under a *Provision Agreement*.

5. Security, privacy, and access

Raw data usually contains direct identifiers and therefore carries the highest confidentiality rating in the chain. Access is limited to the roles that operate the intake and the standardization process. Analysts and subject-matter staff do not work on #1.

Pseudonymisation is normally applied at the transition to #2. Some NSIs pseudonymise at the intake boundary, or receive data already pseudonymised by a trusted third party. Both variants are compatible with the model, provided the choice is documented and the key management is separated from the data. Data minimisation starts before #1: the agreement and the collection design determine that only necessary data is received.

6. Ownership and governance

Role	Responsibility for #1
Data owner	Process owner of Observation/Collection; accountable for lawful acquisition and fitness of the intake
Data steward	Maintains metadata and catalogue entries; resolves gate exceptions
Data custodian	Operates storage, access control, and retention
Relationship manager	Handles provider follow-up on rejected or flagged deliveries (with VS-10)

7. Versioning, retention, and lineage

Every delivery, including a redelivery or correction by the provider, is a new immutable version. A redelivery never overwrites an earlier version, because statistics may already be built on it.

Retention of #1 is governed by two needs that pull in opposite directions. Reproducibility and audit favour keeping raw data; data minimisation and storage cost favour disposal. A common arrangement is to keep #1 until the dependent #3 version is verified and the audit period has passed, while #3 is the state subject to long-term archival requirements. The exact period is set by national law and NSI policy.

Lineage at #1 is the provenance record: the link from the registered version to the provider, the agreement, and the delivery event. Every downstream state must be able to trace back to a #1 version or, where a product starts from another domain's data, to the state it consumed.

8. Reuse and handover

Registration of #1 publishes the *Raw Data Registered* event and a catalogue entry. Consumers discover the delivery through the catalogue and access it through the handover interface, never through direct access to the intake system.

One delivery can serve several statistical products. The register-based census and the labour market statistics may both draw on the same population register delivery. Under SAF-P08, products reuse the registered delivery rather than requesting the same data again from the provider. This also reduces response burden.

Handover element	Content
Artefact	Registered Raw Data version
Required metadata	Minimum metadata set (attribute 4)
Trigger event	<i>Raw Data Registered</i>
Responsibility transfer	VS-01 keeps custody; the consumer takes responsibility for further processing on receipt of the event

9. Traceability to the framework

Framework element	References
Capabilities	1.2.1 Primary data collection, 1.2.2 Secondary data collection, 2.1.1 Data lifecycle management, 2.1.2 Metadata management, 3.3.4 Logical identity and access control, 3.4.3 Service agreement
Value-stream stages	VS-01.3 Collect, Receive, or Extract Data; VS-01.4 Validate and Log Data Intake; VS-01.5 Store and Register Raw Data Artefacts
GSBPM	4.3 Run collection, 4.4 Finalise collection
Principles	SAF-P07, SAF-P08, SAF-P09, SAF-P10, SAF-P11

10. Examples and anti-patterns

Examples

- *Survey*. Responses from a web questionnaire are received with paradata (mode, completion time, instrument version). The delivery is registered as one #1 version per collection batch.
- *Administrative register*. A quarterly tax file arrives by file transfer, with a manifest. Checksum, schema, and record count are checked against the agreement; 14 unknown business identifiers are not a #1 issue and pass through for handling downstream.
- *Web data*. Daily scraped price data is registered with the scraper version and page fingerprint per retailer, so a change in a website's structure can be traced to the affected deliveries.

Anti-patterns

- Correcting obvious errors “while the file is open” at intake. Any value change belongs in #3, where it is documented.
- Overwriting a delivery with the provider's corrected version. Statistics built on the first version lose their trace.
- Giving analysts direct access to raw data for convenience. This bypasses the pseudonymisation at #2.
- Keeping only a converted copy and discarding the original. The proof of authentic custody is lost.
- Registering a delivery without its agreement reference. The legal basis can no longer be demonstrated in an audit.

BE AWARE - State #0 and #1 are updated. The other states will be described the same! TO DO

State #2: Standardized Data

Business Domain Links: Standardization without loss of content

Description: Standardized data has been transformed, formatted, and structured according to defined standards. This is the first point at which interoperability between domains is enabled through common classifications and standardized formats.

Quality Requirements: - Format and structure standardized (e.g., converted from CSV to XML) - Common classifications and code lists applied (e.g., geocoding, economic activity classifications) - Variables mapped to enterprise standards - Reference periods harmonized - Confidentiality measures applied (e.g., pseudonymisation, data minimization) - Data must not contain personally identifiable information

Usage: Standardized data based on the received raw data, enabling data integration and interoperability across the organization.

Key Processing Activities: - Application of reference data and master data - Restructuring and reformatting to NSI's standard storage format - Recoding of variables to align with enterprise standards - Pseudonymisation and encryption where required - Value sets adapted to organizational standards

Interoperability Benefit: This state enables data from different sources to be combined and used across multiple statistical products, reducing duplication and increasing efficiency.

State #3: Processed Data

Business domain Links: Editing, derivation, and non-response correction

Description: Processed data has undergone content-related validation, editing, and enrichment. Data accuracy has been improved through validation checks and corrective measures. This data is ready to be integrated with other datasets.

Quality Requirements: - Completeness and accuracy improved through validation and editing - Data integrity checks completed - Filtering, editing, or imputation applied as needed - Data lineage documented - Integration with other datasets completed where applicable - New derived variables calculated

Usage: Processed data for the processed statistical microdata, ready for statistical analysis and production.

Key Processing Activities: - Data cleansing and editing (content-related standardization) - Integration with other datasets - Validity checking and error correction - Imputation of missing values - Creation of new variables through calculation or derivation - Flagging of problematic values

Domain Responsibility: While this processing is primarily driven by individual statistical domains, depending on the data type, it may be performed by a central unit for efficiency and consistency.

Reusability Focus: Processed data aims to be as reusable as possible for other processes or units, reducing duplication of data and processing effort while increasing transparency.

Long-term retention: Processed data is typically the steady state that national archival requirements apply to: it is the microdata that must be preserved long-term, and without it the statistics built on it cannot be re-verified.

State #4: Output Data (Statistics)

Business Domain Links: Estimation and closed-loop + Statistically composed and composite statistical products

Description: Output data represents statistics that have been aggregated and analyzed, or microdata that has been prepared for specific purposes. Confidentiality measures have been applied to allow potential release.

Quality Requirements: - Data lineage fully documented - Aggregation and analysis completed - Confidentiality measures applied - Statistical estimates calculated - Quality indicators produced - Internal review completed

Usage: Statistics for basic statistics and integrated macrostatistics. These may be used internally by other statistical processes or prepared for dissemination.

Key Characteristics: - Statistics will usually be aggregated data or estimated sizes - May contain confidential and detailed data not yet suitable for publication - Can be shared internally for use as input data in other statistical processes, especially macro statistics - Estimated statistical target quantities produced - Different confidentiality thresholds may apply depending on intended use

Domain-Driven Production: The type of outputs and specific processing methods vary significantly by statistical domain and product requirements.

State #5: Published Output Data (Released Data)

Business Process Links: Disseminate

Description: Published output data has been fully vetted, confidentiality requirements have been satisfied, and the data has been released or published for external use.

Quality Requirements: - Higher confidentiality criteria applied - Stronger documentation requirements met - Data lineage fully traceable - Quality reports completed - Release authorization obtained - Metadata and documentation suitable for external users

Usage: Released data for all data to be disseminated publicly or delivered to authorized external parties.

Key Characteristics: - All requirements for confidentiality taken care of - May be further processed compared to State #4 for additional confidentiality protection - Published by the NSI or delivered to external parties - Includes both published statistical values and delivered microdata (de-identified) - Strict documentation requirements to support user understanding and data reuse

External Consumption: This is the final state in the value chain where statistical information fulfills the core mission of the NSI—delivering trusted statistics to society.

State #6: Consumed Data

Description: Data that has been used by external parties for their own purposes. This represents data outside the direct control of the NSI.

Usage: - None - Consumed data is data used by external data consumers for their analyses and decision-making.

Key Characteristic: This state recognizes that once data is published, it enters the public domain and is used in ways that may be beyond the NSI's direct oversight, though the quality and documentation provided in State #5 support appropriate use.

The Statistical Data Value Chain

The SAF business layer describes how a National Statistical Office (NSI) turns externally-held data into trusted statistics that serve society. Six **primary** value streams form a linear production chain, each adding a specific kind of statistical value, and four **supporting** value streams enable them across the board.

Key concepts — if you are new to architecture documents

This documentation uses a handful of architecture terms. Here is what they mean in plain language:

- **Value stream** — a sequence of steps that together create a statistical output. Think of it as a production line: VS-01 to VS-06 each add a specific kind of value to the data.
- **Value stream stage vs. business process** — the stage (VS-0x.y) says *what value* is added; the business process describes *how* your office actually does it. The numbering maps one-to-one.¹
- **Steady state** — a fixed handover point: a versioned, quality-assured dataset that one stream produces and the next one consumes. The framework defines five numbered steady states, #1 to #5; data still at the external provider (#0) and data in users' hands (#6) sit outside the NSI's control and are boundary positions, not numbered states.
- **The four process types** — every domain is described through the same four groups: **Design** (decide how it should work), **Implementation** (build and test it), **Execution** (run it each production cycle), **Management** (check quality and feed improvements back).
- **Business service / object / event** — what a domain offers to others / the data artefacts it works on / the signals streams send each other (e.g. “*Raw Data Registered*” tells the next stream new data is ready).
- **NSI** — National Statistical Office.
- **GSBPM** — the *Generic Statistical Business Process Model*: the standard, internationally agreed description of the phases of statistical production (specify needs, design, build, collect, process, analyse, disseminate, evaluate).
- **GAMSO** — the *Generic Activity Model for Statistical Organizations*: the companion standard describing management and support activities (strategy, capability development, corporate support).
- **SAF-P01 ... SAF-P11** — the eleven SAF architecture principles, defined in the SAF Framework document and referenced throughout.

This document is the consolidated, easier-to-read view of all ten streams. The shared concepts the primary streams rely on — the chain itself, the steady states of data, the four-process governance pattern, and the

¹This split follows the ArchiMate 3.2 Specification (sec. 7.3.2): “value streams are typically realized by business processes”, with the stages providing “a framework for organizing and defining business processes”.

principles — are explained once below; each **primary** stream is then summarized compactly, keeping its execution detail. The four **supporting** streams follow at the end, in a section that first explains how a supporting stream differs from a production stream. For the full design, implementation, and management detail, and for NSI-application guidance, see the individual VS-0x business-layer documents.

A note on numbering: each VS-0x.y you see below is both a value stream stage (*what value* is added) and a business process (*how* it is done). The numbering maps one-to-one; this document uses the business-process view. See the key-concepts box above for the distinction.

The diagram below shows the whole business layer in one picture: data enters from external providers on the left, gains value through the six primary streams, and reaches users and society on the right — with the four supporting streams enabling every step.

Steady States of Data

The streams work independently of each other (in architecture terms: they are **loosely coupled**). They exchange data only at defined **steady states** — points where a high-quality, versioned dataset is handed over. In the usual case each primary stream consumes one steady state and produces the next, so a stream needs no knowledge of its neighbors’ internals (SAF-P07). The handover works through events: when a stream registers a finished dataset, a notification tells the next stream that new data is ready. Not every process has to pass through every state; see [Applying the steady states pragmatically](#).

Of the five, **processed data (#3) is the only steady state mandatory for all production** — it is the microdata on which everything downstream is verifiably built. **Raw Data (#1)** is also mandatory when a product collects its own data, but not for a macro statistic that starts entirely from another domain’s data. **Standardized data (#2)** is always optional, and **Statistics (#4)** may share its artefact with **Released data (#5)** when the two are identical.

The table lists the five numbered steady states (#1–#5) in the order data passes through them, with the two boundary positions (#0 and #6) outside the NSI’s control shown for context:

Table 1: Steady States of data

State	Name	Produced by	Meaning
(#0)	Source Data	<i>external providers</i>	Data at the external holder, before it enters the NSI (pre-ingestion; not a numbered SAF state)
#1	Raw Data	VS-01	Received into NSI custody in original form, with provenance (a record of where it came from and how) and basic validation
#2	Standardized Data	VS-02	Aligned to common structure, format, and classifications — values unchanged
#3	Processed Data	VS-03	Content validated, edited, imputed, enriched, integrated; derived variables added
#4	Statistics (Output Data)	VS-04 → VS-05	Estimated/aggregated statistics (VS-04), composed into approved products (VS-05)
#5	Released Data	VS-06	Disclosure-controlled, documented, published/delivered to external parties
#6	Consumed Data	<i>external users</i>	Data in use outside the NSI’s direct control

Applying the steady states pragmatically

A steady state is an agreement about data: the quality it has reached, who owns it, how it is identified and what metadata comes with it. It does not require a separate physical copy at every state, and it does not require every process to pass through all six. An office may depart from the usual sequence in the three cases below, provided it records the choice (which state was shared or skipped, and why) in the catalogue entry. In other words, the rule is “comply or explain”.

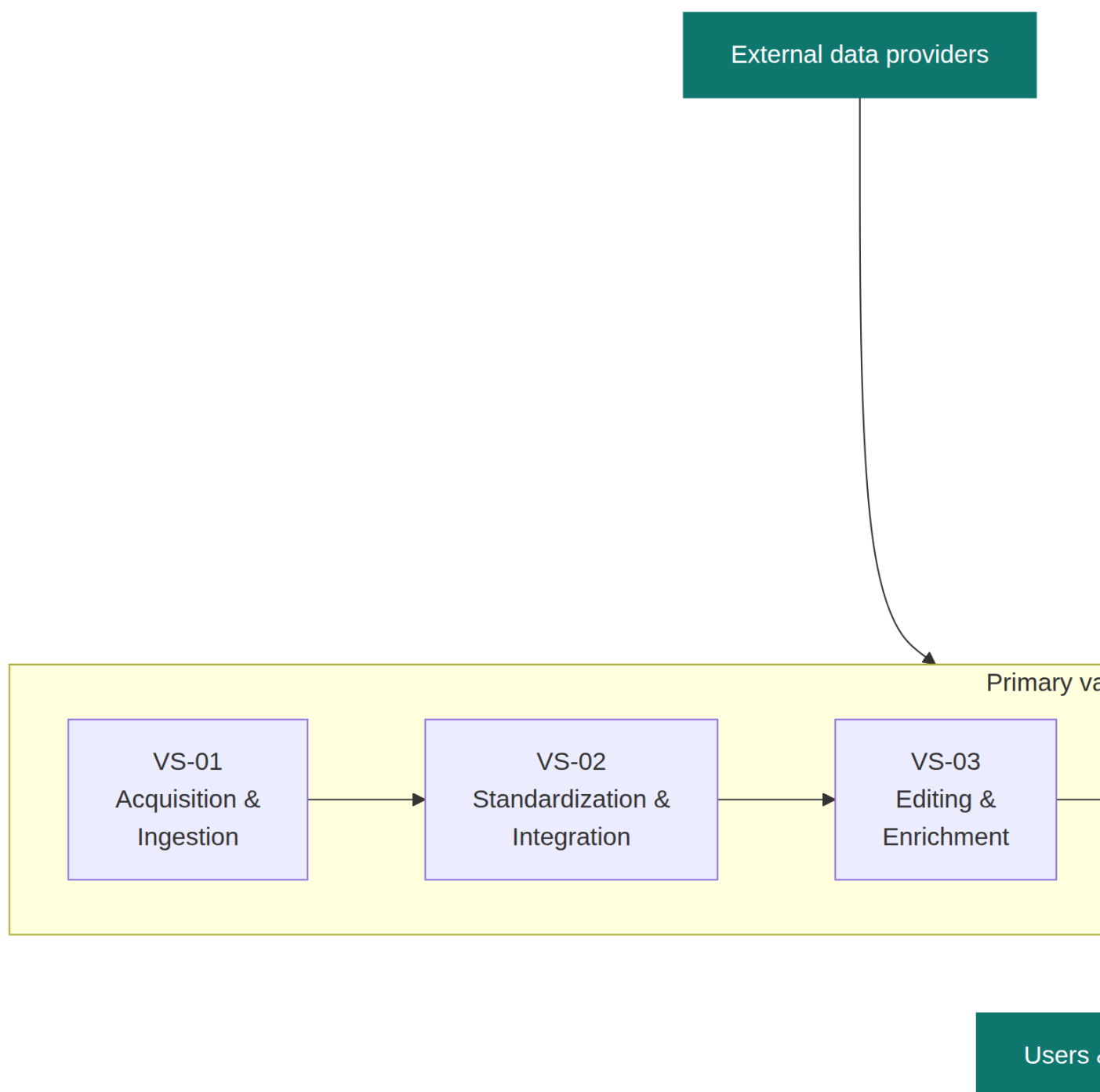
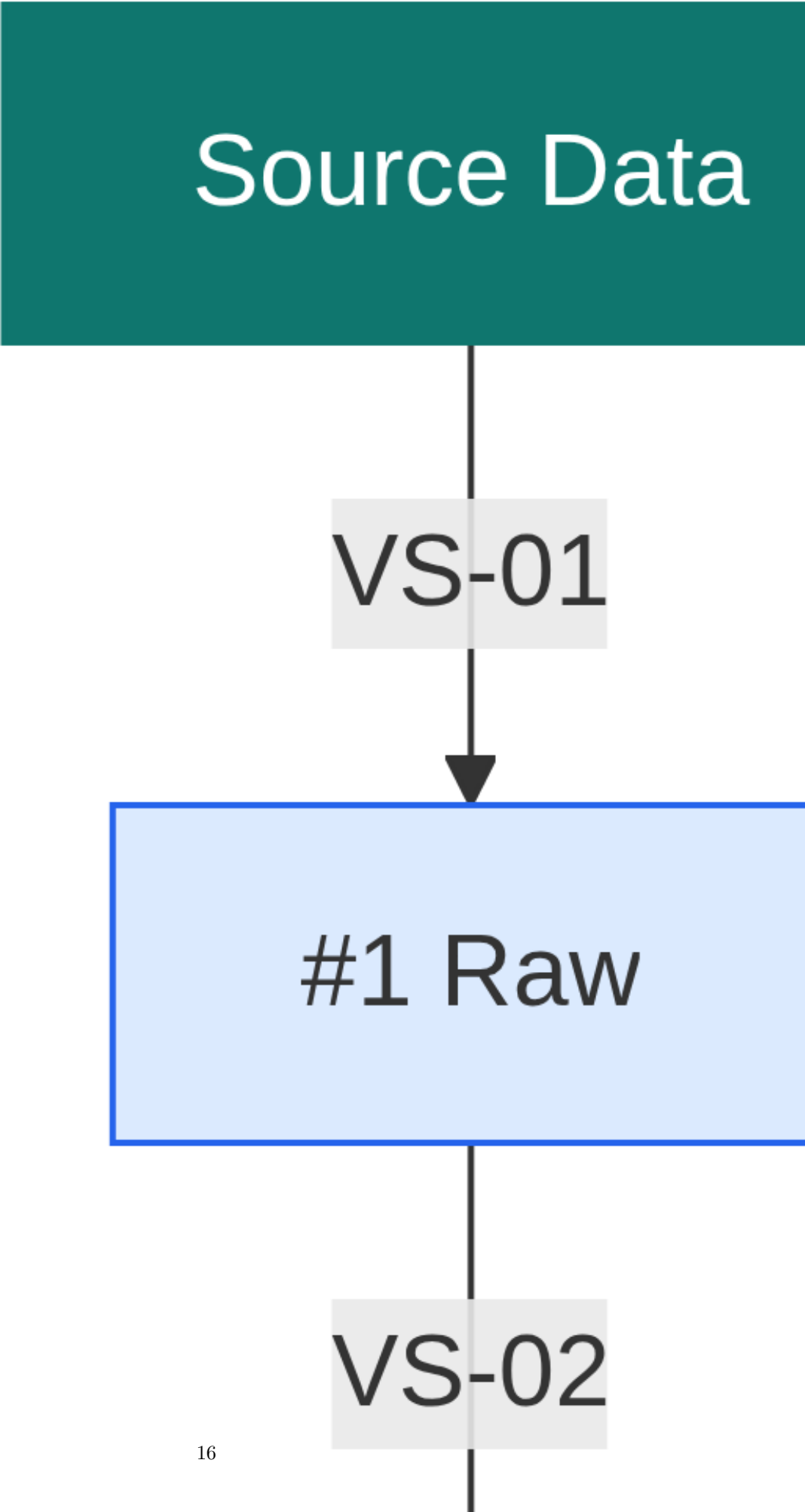


Figure 1: The six primary value streams as a production chain, delivering to users and society. Purple = the four supporting streams (they enable the chain, dashed arrow); dark green = parties outside

Source Data



```
graph TD; A[Source Data] --> B[VS-01]; B --> C[#1 Raw]; C --> D[VS-02];
```

VS-01

#1 Raw

VS-02

1. **Two states can share one artefact.** If nothing happens between State #4 and State #5 (no transformation and no further disclosure control), the #4 artefact does not have to be registered again as a new dataset; it is marked as released instead. The release steps still take place: release authorisation is obtained and the final disclosure check is carried out, and both are recorded even when the check changes nothing.
2. **A process can start downstream.** Not every production process begins at VS-01. Macro statistics, for example, are often built on Processed Data (#3) or Statistics (#4) that another team has already registered, and State #4 above can already be “shared internally for use as input data in other statistical processes”. Such a process starts from the registered artefact and its lineage points to it; there is no need to create placeholder #1 or #2 artefacts. Data that comes from outside the office still enters through VS-01 and State #1.
3. **Lineage is about traceability, not duplication.** SAF-P09 requires that every value can be traced back to where it came from. When a stream uses an artefact without changing any values, a logged process event that refers to the existing artefact version meets this requirement, and no identical copy needs to be registered. As soon as a value changes, a new version is registered.

One limit applies to all three cases. States must not be merged where this would hide a change of values (for example, standardisation at #2 and editing at #3 carried out as a single undocumented step) or a disclosure-control step (for example, #4 and #5 when additional disclosure control is applied before release).

The Four-Process Governance Pattern

Every stream is described through the same **four process types**. This is not a one-off sequence but a repeating cycle that keeps quality improving, and each of the four produces a distinct kind of result:

- **Design** — decide how the work should be done: rules, methodology, specifications, policies.
- **Implementation** — build, configure, and test the systems that will do it.
- **Execution** — run the work each production cycle; this is what produces the statistical data.
- **Management** — check quality, monitor performance, and feed improvements back to Design.

The cycle looks like this — note the feedback arrow that closes the loop:

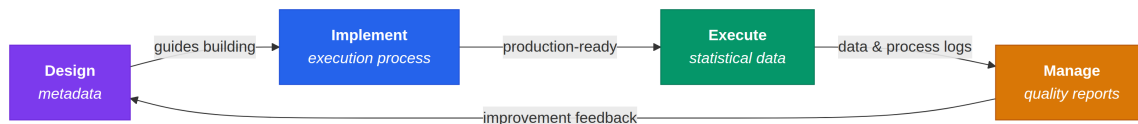


Figure 3: The four-process governance pattern. Each of the six streams runs this full cycle; the per-stream sections below summarize Design/Implementation/Management and keep the Execution detail.

i Note

In the sections that follow, GSBPM phases are mapped per process type:

- **Design** → **Phase 2**,
- **Implementation** → **Phase 3**,
- **Execution** → **the later GSBPM phases** appropriate to the domain (Phase 4 for collection, Phase 5 for processing, Phase 6 for analysis, Phase 7 for dissemination),
- **Management** → **overarching Quality Management / Phase 8**.

The four **supporting** streams (VS-07–VS-10) do not follow this production cadence: they run a **continuous plan → build → operate → improve** cycle and align to **GAMSO** rather than GSBPM (see the supporting-streams section).

Principles to Apply

The SAF defines eleven architecture principles, labeled SAF-P01 to SAF-P11. The six **primary** streams actively apply six of them (marked below); the four **supporting** streams foreground the other five (marked †). The table shows each principle with practical guidance on how to apply it; VS-07 in particular **enforces** the data principles (P09/P10/P11) across the whole chain.

Table 2: SAF Principles

Principle	Statement	How to apply across the value chain
SAF-P01	Actively participate in open source to maintain collective control over statistical processes	Build statistical capabilities on shared, open tooling; contribute back, so NSIs keep collective control of critical methods rather than depending on vendors.
SAF-P07	Implement loosely coupled processes and systems	Exchange data only at steady-state interfaces; each stream consumes one state and produces the next without knowing the others' internals.
SAF-P08	Data is shared property	Once an artefact is registered at a steady state, make it available to all authorized consumers; avoid duplicating collection, processing, or analysis.
SAF-P09	No data without metadata and classification	Every artefact carries provenance (origin), methodology, accuracy, classification, and lineage (the trace back to its inputs) — captured at the point each value is created or changed.
SAF-P10	Security By Design	Embed access control, integrity verification, and tamper-evident audit trails in every pipeline from the start.
SAF-P11	Privacy By Design	Remove/pseudonymize identifiers during standardization; assess disclosure risk on enrichment and apply statistical disclosure control before release.
SAF-P02 †	Active support	Operate and support capabilities as living services, not one-off builds (VS-09 platforms, VS-10 services).
SAF-P03 †	Active Life Cycle management	Version and govern rules, methods, models, and products across their life cycle; retire what is obsolete (VS-07, VS-08).
SAF-P04 †	Maximize the benefits for the organization	Prefer solutions that serve the whole organization and real user value over local optimizations (VS-10, VS-07).
SAF-P05 †	IT and technology is everyone's business	Treat methodology, data, and technology as a shared responsibility across business and IT (VS-08, VS-09).
SAF-P06 †	Business Continuity	Keep the chain running reliably and able to recover from disruption (VS-09).

= actively applied by the six primary value streams. † = foregrounded by the four supporting value streams (VS-07–VS-10).

! Two cross-chain disciplines

Metadata everywhere (SAF-P09): value is only added once its context is captured — what was done, by which rule/method, with what accuracy, and traceable to its inputs.

Confidentiality progressively applied (SAF-P11): identifiers are removed/pseudonymized at standardization (#2) at the latest, or earlier where that suits the process; disclosure risk is assessed on enrichment (#3), disclosure control is applied to outputs (#4), and a final disclosure check gates release (#5).

VS-01.0 — Data Acquisition and Ingestion

Domain: Observation/Collection · **State transition:** Source Data → #1 Raw Data · **Value:** all data enters NSI custody, verifiably and with provenance.

VS-01 is the front door of the statistical system. Its job is to bring data from the outside world — survey respondents, administrative registers, and emerging sources such as web or sensor data — into the NSI safely and with a complete record of where it came from. Nothing about the data’s content is changed here; the point is to receive it, prove it is what it claims to be, and register it so the rest of the chain can find and trust it. The output is **Raw Data**: the data in its original form, under NSI custody, accompanied by provenance.

Business processes

- VS-01.1 Identify Data Needs and Sources
- VS-01.2 Establish Data Agreement and Access Right
- VS-01.3 Collect, Receive, or Extract Data
- VS-01.4 Validate and Log Data Intake
- VS-01.5 Store and Register Raw Data Artefacts

How the business processes work. The cycle starts by **identifying the data needs and sources** for the round — confirming what variables and populations are required and that the chosen sources are available and accessible. Once the source is settled, the team **establishes the data agreement and access rights**: activating or renewing the legal basis (a data-sharing agreement, or the relevant GDPR ground) and checking that credentials are valid, so collection is lawful and controlled. Data is then **collected, received, or extracted** through the appropriate channel — a survey instrument, an administrative file transfer, or an API/scraping job — and at the moment of receipt the ingestion metadata (source, method, timestamp, file hash, volume) is captured. Each delivery is **validated and logged**: completeness, structure, and integrity are checked and every result written to an audit trail, so the NSI can be confident the intake is whole and traceable. Finally the dataset is **stored and registered as a raw-data artefact** — given a persistent identifier, kept in its original form, and cataloged with its provenance — and a “Raw Data Registered” event tells VS-02 that new data is ready.

i Signature discipline — no data without metadata

Every dataset is enriched at ingestion with provenance, source, method, timestamp, and a quality and BIV classification — availability, integrity, confidentiality — per SAF-P09, with security and privacy applied by design (SAF-P10/P11).

Around this per-cycle run, **Design** prepares the data requirements, collection methodology, raw-data schema, validation rules, and access/privacy framework; **Implementation** builds the collection instruments and ingestion/validation workflows and tests them; and **Management** monitors response, timeliness, and rejection rates, assesses intake quality, and feeds improvements back to design.

Business process	What it produces
VS-01.1 Identify Data Needs and Sources	Confirmed requirements and available sources for the cycle
VS-01.2 Establish Data Agreement and Access Right	Active, lawful access (agreements, GDPR basis, credentials)
VS-01.3 Collect, Receive, or Extract Data	Data in NSI custody + captured ingestion metadata
VS-01.4 Validate and Log Data Intake	Validated intake + audit trail
VS-01.5 Store and Register Raw Data Artefacts	Registered Raw Data artefact + “Raw Data Registered” event

Boundaries: in — Source Data from external providers; out — **Raw Data** on “Raw Data Registered” → VS-02.

VS-02.0 — Data Standardization and Integration

Domain: Standardization without loss of content · **State transition:** #1 Raw → #2 Standardized · **Value:** coherence and comparability across sources.

VS-02 makes data from different sources speak the same language. After raw data is in custody, this stream aligns its structure, format, and codes to the NSI’s agreed standards so that datasets which looked different at the source can be compared and combined. Crucially it does this **without changing any value** — it re-expresses and annotates, it does not transform. The output is **Standardized Data**: the same information, now in a common shape, expressed in standard classifications, and carrying rich metadata.

Business processes

- VS-02.1 Profile and Assess Data Structure
- VS-02.2 Apply Standard Classifications and Formats
- VS-02.3 Align Metadata and Semantics
- VS-02.4 Validate Structural and Semantic Conformance
- VS-02.5 Register Standardized Artefacts

How the business processes work. The stream first **profiles and assesses the incoming structure**, comparing it against the target data model to see which fields map directly, which simply need reformatting, which source codes have a standard correspondence, and which values are uncoded. It then **applies standard classifications and formats**: source codes are mapped one-to-one to standards (for example a source activity code to NACE, the European industry classification) through maintained correspondence tables, and the layout is coerced into the common format (renaming, restructuring, type-casting, e.g. CSV→XML). The original values are preserved, and anything that has no deterministic correspondence — or is free text needing interpretation — is flagged and passed downstream to VS-03. Next it **aligns metadata and semantics**: variables are linked to canonical concepts, and each variable’s unit, currency, and reference-period basis (and any differing cross-source definitions) are *recorded as metadata* rather than converted. The result is **checked for structural and semantic conformance**, including a no-loss-of-content check confirming that record counts and key measures are unchanged. Conformant data is then **registered as a standardized artefact** with its mapping lineage, and a “Standardized Data Registered” event notifies VS-03.

i Signature discipline — standardization without loss of content

VS-02 only maps codes 1-to-1, reformats structure, and annotates metadata. It **never changes a value**; unit/currency conversion, reference-period alignment, value reconciliation, and interpretive free-text coding are deliberately deferred to VS-03.

Around this, **Design** defines the target data model, the deterministic classification correspondences, the harmonization and conformance rules, and the privacy framework; **Implementation** builds the mapping & format-coercion component, the classification-mapping services, the conformance pipeline, and identifier pseudonymization; **Management** tracks mapping coverage and the unmapped-value rate, governs classification versions, and improves the correspondences.

Business process	What it produces
VS-02.1 Profile and Assess Data Structure	A structure-gap assessment against the target model
VS-02.2 Apply Standard Classifications and Formats	Standard-coded, common-format data (values unchanged); unmapped values flagged for VS-03
VS-02.3 Align Metadata and Semantics	Canonical-concept links + unit/period/definition metadata
VS-02.4 Validate Structural and Semantic Conformance	Conformance result + no-loss-of-content confirmation
VS-02.5 Register Standardized Artefacts	Registered Standardized Data + “Standardized Data Registered” event

Boundaries: in — Raw Data on “*Raw Data Registered*”; out — **Standardized Data** on “*Standardized Data Registered*” → VS-03.

VS-03.0 — Data Editing and Enrichment

Domain: Editing, derivation, and non-response correction · **State transition:** #2 Standardized → #3 Processed · **Value:** data reliability and analytical readiness.

VS-03 is where data quality is actively improved. Standardized data is rarely clean — it contains errors, inconsistencies, gaps, and not-yet-coded values. This stream detects those problems, corrects them, fills the gaps, and derives the extra variables analysis needs. Uniquely in the chain, it **deliberately changes values to do so**. What makes that safe is total traceability: every change is recorded against its original value and a reproducible, versioned dataset is produced. The output is **Processed Data**: reliable, complete, analytically ready microdata.

Business processes

- VS-03.1 Conduct Error Detection / Quality Diagnostics
- VS-03.2 Perform Data Editing and Correction
- VS-03.3 Impute Missing or Non-Response Data
- VS-03.4 Derive Additional Variables and Indicators
- VS-03.5 Perform Quality Evaluation and Versioning

How the business processes work. The stream begins by **detecting errors and running quality diagnostics** — consistency, range, outlier, and duplicate checks — and uses selective-editing scores to prioritize the problems that actually matter for the results. It then **performs editing and correction**: low-impact issues are auto-corrected by rule while high-impact records are reviewed by subject-matter analysts, and this is where the interpretive work VS-02 deferred is carried out (coding free-text values such as occupation to ISCO, the international occupation classification, and reconciling values where integrated sources disagree). Every change logs the original value, the rule or decision, the reason, the actor, and the time. Genuine gaps are then closed by **imputing missing or non-response data** using approved methods, with each imputed value clearly flagged so it stays distinguishable from observed data. The dataset is **enriched by deriving additional variables and indicators** — and this is also where unit/currency conversion and reference-period alignment are computed from the bases VS-02 recorded, and where datasets are integrated on common keys. Finally the stream **evaluates quality and versions the result**: it measures the improvement against the incoming baseline, verifies the edit lineage is complete and reversible, and creates an immutable Processed Data version, signaled to VS-04 by a “Processed Data Registered” event.

i Signature discipline — full traceability

VS-03 is the domain that changes values, under the guarantee of *no undocumented or irreversible change*: original values are preserved beside edited ones, imputations are flagged, and every version is reproducible.

Around this, **Design** sets the error-detection and editing rules (with a selective-editing strategy), the imputation methodology, the derivation and coding specs, the integration rules, and the versioning policy; **Implementation** builds the editing, imputation, coding, derivation, and integration engines and the version store; **Management** watches edit, imputation, and over-editing rates, assesses quality improvement, and audits the edit lineage.

Business process	What it produces
VS-03.1 Conduct Error Detection / Quality Diagnostics	A prioritized diagnosis of errors and gaps
VS-03.2 Perform Data Editing and Correction	Corrected values + interpretive coding + full edit lineage
VS-03.3 Impute Missing or Non-Response Data	Completed data with imputation flags
VS-03.4 Derive Additional Variables and Indicators	Derived variables + unit/currency/period conversions + integrated data
VS-03.5 Perform Quality Evaluation and Versioning	Versioned Processed Data + “Processed Data Registered” event

Boundaries: in — Standardized Data on “*Standardized Data Registered*”; out — **Processed Data** on “*Processed Data Registered*” → VS-04.

VS-04.0 — Estimation, Analysis, and Statistical Modeling

Domain: Estimation and closed-loop · **State transition:** #3 Processed → #4 Statistics · **Value:** analytical insight and statistical credibility.

VS-04 turns clean microdata into statistics. Up to now the chain has improved the data itself; here the focus shifts to producing numbers that describe a population — estimates, weighted aggregates, indices, and model results. The microdata is not edited further; instead statistics are computed *from* it, each accompanied by a measure of how accurate it is. The output is **Output Data / Statistics**: validated figures with their methodology and accuracy attached, and an initial layer of confidentiality protection applied.

Business processes

- VS-04.1 Confirm Estimation and Analytical Approach
- VS-04.2 Apply Weighting and Calibration
- VS-04.3 Estimate and Aggregate Data
- VS-04.4 Perform Model-Based Analysis
- VS-04.5 Validate and Document Results

How the business processes work. Each round starts by **confirming the estimation and analytical approach** — selecting the pre-approved method for this cycle’s data and checking that the calibration totals and domains are current. The stream then **applies weighting and calibration** to correct for sampling, non-response, and coverage, so the sample represents the population. With weights in place it **estimates and aggregates the data**, producing the target statistics together with their accuracy (standard errors and confidence intervals) — and because VS-03 flagged which values were imputed, that added uncertainty is reflected in the variance. Where direct figures are not enough, it **performs model-based analysis** (regression, time-series/forecasting, small-area models) to reveal relationships and produce reliable estimates for thin domains, recording the model diagnostics. Finally it **validates and documents the results**: checking statistical soundness and coherence with related and prior statistics, applying an initial layer of statistical disclosure control so the figures can potentially be released, documenting the methodology, and versioning the output — announced to VS-05 by an “Output Data Registered” event.

i Signature discipline — methodological transparency

Every estimate carries the method, parameters, and accuracy measure that produced it and can be regenerated; imputation uncertainty from VS-03 is reflected in the variance, and initial statistical disclosure control is applied so output can potentially be released.

Around this, **Design** fixes the estimation strategy and accuracy approach, the weighting/calibration methodology, the aggregation and model specifications, and the validation and disclosure-control policy; **Implementation** builds the weighting, estimation, variance, and modeling engines and the disclosure-control and version tooling; **Management** tracks accuracy, timeliness, and revisions, assesses coherence, and audits reproducibility.

Business process	What it produces
VS-04.1 Confirm Estimation and Analytical Approach	The pre-approved method confirmed for the cycle
VS-04.2 Apply Weighting and Calibration	Weights that make the sample represent the population
VS-04.3 Estimate and Aggregate Data	Estimates and aggregates, each with an accuracy measure
VS-04.4 Perform Model-Based Analysis	Model/forecast/small-area results with diagnostics
VS-04.5 Validate and Document Results	Validated, disclosure-controlled, versioned statistics + “Output Data Registered” event

Boundaries: in — Processed Data on “*Processed Data Registered*”; out — **Output Data** on “*Output Data Registered*” → VS-05.

VS-05.0 — Statistical Product Composition

Domain: Statistically composed and composite products · **State transition:** within #4 **Statistics** (analytical outputs → approved products) · **Value:** coherence and interpretability.

VS-05 assembles finished products from the statistics VS-04 produced. Individual estimates are not yet something a user reads; this stream brings results from different domains, surveys, and time periods together into coherent publications, time series, indices, and datasets — and approves them for release. It does not re-estimate anything; it selects, combines, checks, explains, and signs off. The output is an **approved, registered statistical product** ready for dissemination.

Business processes

- VS-05.1 Select and Integrate Source Datasets
- VS-05.2 Construct Composite Indicators or Indices
- VS-05.3 Perform Consistency and Coherence Checks
- VS-05.4 Generate Analytical and Interpretive Outputs
- VS-05.5 Approve and Register Statistical Products

How the business processes work. The stream first **selects and integrates the source datasets** — picking the exact validated output versions a product needs and combining them, carrying their accuracy metadata along. It then **constructs the composite indicators or indices** the product presents (ratios, indices, chained time series), deriving each composite’s uncertainty from its components. The assembled product is put through **consistency and coherence checks**: totals must reconcile to their parts, and the figures must agree with related statistics and with prior releases — discrepancies are explained or sent back, never quietly adjusted. Next the stream **generates the analytical and interpretive outputs** — the commentary, key messages, tables, and visualizations that make the product understandable — and runs them through editorial review. Finally it **approves and registers the product**: validating it against dissemination and confidentiality standards, obtaining sign-off from the designated authority, and registering an immutable version with full lineage, which a “Statistical Product Registered” event passes to VS-06.

i Signature discipline — coherence and interpretability

VS-05 **composes** validated outputs (it does not re-estimate). Products are internally consistent, coherent with related and prior statistics, carry their components’ accuracy metadata, and are **approved against dissemination and confidentiality standards** before being registered.

Around this, **Design** defines the product portfolio, the composite-indicator specs, the coherence checks, the content templates, and the approval/confidentiality/versioning framework; **Implementation** builds the integration and composition engines, the indicator/time-series engines, the coherence and content tooling, and the approval/version store; **Management** tracks coherence-check pass rates and approval iterations, assesses product quality, and audits reproducibility and approval.

Business process	What it produces
VS-05.1 Select and Integrate Source Datasets	Integrated components with accuracy carried forward
VS-05.2 Construct Composite Indicators or Indices	Composite measures and time series
VS-05.3 Perform Consistency and Coherence Checks	Confirmed internal consistency and coherence
VS-05.4 Generate Analytical and Interpretive Outputs	Reviewed narrative, tables, and visualizations
VS-05.5 Approve and Register Statistical Products	Approved, versioned product + “Statistical Product Registered” event

Boundaries: in — Output Data on “*Output Data Registered*”; out — **Approved Statistical Product** on “*Statistical Product Registered*” → VS-06.

VS-06.0 — Dissemination and Service Delivery

Domain: Dissemination (final stage) · **State transition:** #4 Approved products → #5 Released → #6 Consumed · **Value:** impact and accessibility.

VS-06 is the chain’s exit point — where approved products become public value. It publishes and delivers statistics to policymakers, researchers, journalists, businesses, and the public through trusted, secure channels, and supports users in interpreting them correctly. The decisive control here is confidentiality: a final disclosure-control gate stands between an approved product and release. The output is **Released Data** (State #5); once in users’ hands it becomes **Consumed Data** (State #6), beyond the NSI’s direct control.

Business processes

- VS-06.1 Plan Dissemination and Access Strategy
- VS-06.2 Prepare Data for Dissemination
- VS-06.3 Publish and Provide Access Services
- VS-06.4 Support and Engage Users
- VS-06.5 Monitor Usage and Feedback

How the business processes work. Each release begins by **confirming the dissemination and access plan** — the channels (publication, API/open data, dashboards, secure research access), the formats, the release-calendar slot, and the embargo/equal-access arrangements. The product is then **prepared for dissemination**: final statistical disclosure control is applied (suppression, rounding, perturbation; de-identification of any microdata) and the release must clear the disclosure-risk criteria — *nothing proceeds if it does not* — after which it is formatted for each channel and packaged with its documentation and metadata. The stream then **publishes and provides access services**, releasing the product and its metadata across the confirmed channels with timing managed so that access is equal (no privileged early sight outside policy). It **supports and engages users** with documentation, a helpdesk, training, and errata, so the statistics are interpreted responsibly. Finally it **monitors usage and feedback** — tracking downloads, API calls, and dashboard views, and collecting satisfaction and questions — and routes those insights back to design, to VS-10, and across the chain. The release is signaled by a “Statistics Released” event.

i Signature discipline — accessibility, trust, and the release gate

Accurate, timely, understandable delivery through secure channels, gated by **final statistical disclosure control** (no release without confidentiality clearance) and supported by documentation that enables responsible, equal-access use. VS-06 relies on the supporting streams VS-07 (compliance), VS-09 (platforms), and VS-10 (user needs).

Around this, **Design** sets the dissemination and access strategy, the final disclosure-control and de-identification rules, the channels/formats/documentation standards, and the release governance; **Implementation** builds the publication, API, dashboard, and secure-research-access services (on VS-09 platforms) and the disclosure-control and release tooling; **Management** monitors punctuality, usage, and satisfaction, assesses accessibility, and audits disclosure and equal-access compliance with VS-07.

Business process	What it produces
VS-06.1 Plan Dissemination and Access Strategy	Confirmed channels, formats, and release timing
VS-06.2 Prepare Data for Dissemination	Disclosure-safe, channel-ready products + documentation
VS-06.3 Publish and Provide Access Services	Released statistics across channels, equally and on time
VS-06.4 Support and Engage Users	User documentation, helpdesk, training, errata
VS-06.5 Monitor Usage and Feedback	Usage insight + feedback routed to design and VS-10

Boundaries: in — Approved Statistical Product on “*Statistical Product Registered*”; out — **Released Data** on “*Statistics Released*” → users/society (consumed as State #6). No downstream primary stream; feedback returns to VS-10.

Supporting Value Streams (VS-07 to VS-10)

The four supporting streams are not stages in the production line — they work **across the whole chain**, enabling and constraining all six primary streams continuously. Reading their sections requires a small shift of mindset from the six above.

! How a supporting value stream differs from a production stream

- **No steady-state data transition.** A supporting stream does not take data from one state to the next; it does not appear in the #1→#6 chain. There is no “Data Registered” handover event and no “Boundaries: in→out” — instead each interacts with the whole chain at once.
- **It runs continuously**, as a plan → build → operate → improve cycle, not per production cycle.
- **Its output is an enabling capability, not data:** governance & assurance (07), validated methods (08), platforms (09), understood needs & relationships (10).
- **It aligns to GAMSO** — the standard model of management and support activities (Strategy & Leadership, Capability Management, Corporate Support) — rather than to GSBPM production phases, and it foregrounds principles **P02–P06**.

The diagram shows the two directions of that interaction: the supporting streams provide capabilities to the chain (arrows in), and the chain sends back evidence, needs, and usage (arrow out):

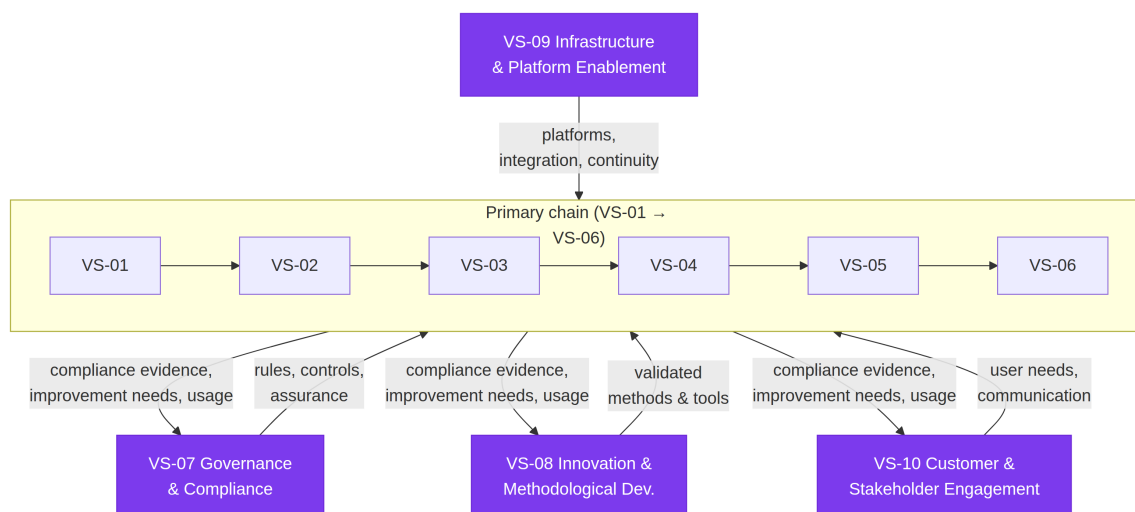


Figure 4: The four supporting streams (purple) and the primary chain: each provides a capability to the chain and receives feedback or needs from it.

VS-07.0 — Governance and Compliance

Domain: Governance (horizontal) · **GAMSO:** Strategy & Leadership + Corporate Support · **Value:** integrity and accountability.

VS-07 makes sure every process, dataset, and product operates under clear rules, standards, and accountability. It builds trust by ensuring all data activities are ethical, transparent, and compliant with legal, methodological, and quality requirements (Code of Practice, GDPR, ISO). It does not move data along the chain — it **oversees** the streams that do, and is the place where the data principles (P09/P10/P11) are enforced organization-wide.

Business processes

- VS-07.1 Define Governance Frameworks and Policies
- VS-07.2 Establish Legal and Regulatory Compliance Mechanisms
- VS-07.3 Implement Data and Metadata Governance
- VS-07.4 Monitor Quality and Compliance Indicators
- VS-07.5 Audit, Certify, and Report Compliance
- VS-07.6 Update and Improve Governance Mechanisms

How the business processes work. Governance runs as a continuous loop. It first **defines governance frameworks and policies** (07.1) — the charters and principles covering data, metadata, ethics, confidentiality, and quality — and **establishes legal and regulatory compliance mechanisms** (07.2), mapping applicable regulation to concrete procedures. These are made operational by **implementing data and metadata governance** (07.3): assigning the owner/steward/custodian roles and the metadata and lineage processes that wire governance into how the streams work. It then **monitors quality and compliance indicators** (07.4) on dashboards and periodically **audits, certifies, and reports compliance** (07.5), producing the assurance that the statistics can be trusted. Finally it **updates and improves the governance mechanisms** (07.6), folding regulatory change and lessons learned back into the policies. *(The strategy model labels two steps VS-07.5; the second is treated here as VS-07.6.)*

Interaction with the chain: provides frameworks, roles, and controls to all streams and receives their compliance evidence, quality indicators, and audit findings (each primary stream’s audit-and-compliance reporting feeds here); issues certification and assurance to regulators and the public.

Business process	What it produces
VS-07.1 Define Governance Frameworks and Policies	Governance charters and policies
VS-07.2 Establish Legal and Regulatory Compliance Mechanisms	Compliance procedures mapped to regulation
VS-07.3 Implement Data and Metadata Governance	Assigned roles + metadata/lineage processes
VS-07.4 Monitor Quality and Compliance Indicators	Quality & compliance dashboards
VS-07.5 Audit, Certify, and Report Compliance	Audit reports, certifications, assurance
VS-07.6 Update and Improve Governance Mechanisms	Revised policies and controls

VS-08.0 — Innovation and Methodological Development

Domain: Innovation (horizontal) · **GAMSO:** Capability Management · **Value:** continuous improvement / future-proofing.

VS-08 keeps the organization from standing still: it discovers, tests, and introduces new data sources, methods, tools, and technologies, connecting research, methodology, IT, and business users. It keeps innovation **controlled** so that only sound, documented, compliant methods reach production. It does not transform data — it improves the methods and tools the production streams use.

Business processes

- VS-08.1 Identify Innovation Needs and Opportunities
- VS-08.2 Conduct Research and Proof-of-Concepts
- VS-08.3 Evaluate Methodological Soundness and Feasibility
- VS-08.4 Develop / Document New Methodologies or Tools
- VS-08.5 Integrate Innovation into Production Environments
- VS-08.6 Promote Knowledge Sharing and Capability Building

How the business processes work. The pipeline begins by **identifying innovation needs and opportunities** (08.1) — scanning technology and methodology (AI, ML, new sources) and the chain’s own improvement backlog — then runs **research and proofs-of-concept** (08.2) in limited scope. Promising results pass through **evaluation of methodological soundness and feasibility** (08.3), the gate that stops attractive-but-unsound ideas, and are then **developed and documented into methods or tools** (08.4) — standardized procedures and reference implementations ready to hand over. The defining step is **integrating innovation into production** (08.5): the governed transfer of a validated method into the relevant primary stream (after VS-07 approval, on VS-09 platforms). Finally it **promotes knowledge**

sharing and capability building (08.6), disseminating guidance and growing staff competency — which surfaces the next round of needs.

Interaction with the chain: receives improvement needs from every primary stream (their continuous-improvement loops) and user-driven needs from VS-10; **delivers validated methods and tools into production** through the 08.5 transfer gate.

Business process	What it produces
VS-08.1 Identify Innovation Needs and Opportunities	Prioritized innovation backlog
VS-08.2 Conduct Research and Proof-of-Concepts	Experiment / pilot results
VS-08.3 Evaluate Methodological Soundness and Feasibility	Soundness & feasibility assessment
VS-08.4 Develop / Document New Methodologies or Tools	Standardized method + reference implementation
VS-08.5 Integrate Innovation into Production Environments	Method/tool live in a production stream (governed transfer)
VS-08.6 Promote Knowledge Sharing and Capability Building	Disseminated guidance + trained staff

VS-09.0 — Infrastructure and Platform Enablement

Domain: Infrastructure (horizontal) · **GAMSO:** Capability Management + Corporate Support · **Value:** reliable enablement at scale.

VS-09 provides the technical and digital foundation every other stream runs on — environments, data platforms, tools, and integration services — keeping them secure, resilient, and compliant while flexible enough to adopt new technology. It is the business-layer view of providing the application and technology enablement the primary documents scoped out.

Business processes

- VS-09.1 Define Digital and Technical Requirements
- VS-09.2 Design Platform and Infrastructure Architecture
- VS-09.3 Provision and Configure Platforms
- VS-09.4 Enable Integration and Orchestration Services
- VS-09.5 Ensure Performance, Security, and Continuity
- VS-09.6 Evolve and Modernize the Technical Ecosystem

How the business processes work. The lifecycle begins by **defining digital and technical requirements** (09.1) from business capabilities and data volumes, feeding the **design of the platform and infrastructure architecture** (09.2) — blueprints for storage, processing, integration, and secure access. The architecture is realized by **provisioning and configuring platforms** (09.3) with governance, monitoring, and security controls, and by **enabling integration and orchestration services** (09.4) — the APIs, workflow engines, and the event interfaces the steady-state handovers rely on. Day to day it **ensures performance, security, and continuity** (09.5): monitoring, data protection, incident management, and disaster recovery. Finally it **evolves and modernizes the technical ecosystem** (09.6), adopting new technology (often from VS-08) and retiring the obsolete.

Interaction with the chain: provides environments, platforms, integration, and assured availability to all streams (including VS-06's portals/APIs and VS-08's sandboxes); receives requirements, capacity demand, and incidents; its security and continuity posture is governed by VS-07.

Business process	What it produces
VS-09.1 Define Digital and Technical Requirements	Infrastructure requirements from business needs
VS-09.2 Design Platform and Infrastructure Architecture	Architecture blueprints
VS-09.3 Provision and Configure Platforms	Deployed, controlled environments

Business process	What it produces
VS-09.4 Enable Integration and Orchestration Services	APIs, workflow/automation, event interfaces
VS-09.5 Ensure Performance, Security, and Continuity	Monitoring, security, availability, DR
VS-09.6 Evolve and Modernize the Technical Ecosystem	Modernization roadmap and adopted technology

VS-10.0 — Customer and Stakeholder Engagement

Domain: Engagement (horizontal) · **GAMSO:** Strategy & Leadership + Corporate Support · **Value:** relevance and public value.

VS-10 keeps the organization connected, relevant, and responsive — building mutual understanding and trust with users, data providers, partners, and policymakers. It is a continuous two-way dialogue: it does not move data, it moves **understanding** — needs in, communication and services out.

Business processes

- VS-10.1 Identify and Segment Stakeholders
- VS-10.2 Collect and Analyze User Needs
- VS-10.3 Define and Manage Service Portfolios
- VS-10.4 Deliver Services and Manage Relationships
- VS-10.5 Engage and Communicate with Stakeholders
- VS-10.6 Monitor Impact and Satisfaction

How the business processes work. The cycle begins by **identifying and segmenting stakeholders** (10.1) and **collecting and analyzing user needs** (10.2), translating them into data or service requirements. Those needs shape the **definition and management of service portfolios** (10.3) — catalogs of services and products aligned to strategy. Services reach users through **delivering services and managing relationships** (10.4) — support desks, account management, satisfaction monitoring — alongside **engaging and communicating with stakeholders** (10.5) via outreach and knowledge-sharing. Finally it **monitors impact and satisfaction** (10.6), measuring use and societal impact and feeding insight back into planning and innovation.

Interaction with the chain: bidirectional with VS-01 (the data providers it engages) and VS-06 (whose usage and feedback it receives); it routes user needs into design across the primary streams and into VS-08's innovation backlog.

Business process	What it produces
VS-10.1 Identify and Segment Stakeholders	Stakeholder map and segments
VS-10.2 Collect and Analyze User Needs	Needs register / service requirements
VS-10.3 Define and Manage Service Portfolios	Service & product portfolio
VS-10.4 Deliver Services and Manage Relationships	Operating support & managed relationships
VS-10.5 Engage and Communicate with Stakeholders	Outreach, communication, awareness
VS-10.6 Monitor Impact and Satisfaction	Impact & satisfaction insight (fed back to the chain)

The Value Chain at a Glance

Use these two tables as a quick reference: the first summarizes the six production streams, the second the four supporting streams.

VS	Domain	State in → out	Trigger out	Signature discipline
01	Acquisition & Ingestion	Source → #1 Raw	Raw Data Registered	Metadata & validation at ingestion
02	Standardization & Integration	#1 → #2 Standardized	Standardized Data Registered	No loss of content: map, reformat, annotate
03	Editing & Enrichment	#2 → #3 Processed	Processed Data Registered	Full traceability of every value change
04	Estimation & Modeling	#3 → #4 Statistics	Output Data Registered	Methodological transparency + accuracy + disclosure control on outputs
05	Product Composition	within #4 (→ approved)	Statistical Product Registered	Coherence & interpretability; approval before registration
06	Dissemination & Service Delivery	#4 → #5 Released → #6	Statistics Released	Accessibility & trust; final disclosure-control gate

The four supporting streams work differently — they have no state transition; they run horizontally and continuously:

VS	Domain	GAMSO area	What it provides to the chain	Principles foregrounded
07	Governance & Compliance	Strategy & Leadership; Corporate Support	Frameworks, controls, assurance; enforces the data principles	P03, P04 (+ enforces P09/P10/P11)
08	Innovation & Methodological Development	Capability Management	Validated methods & tools transferred into production	P01, P05, P03, P04
09	Infrastructure & Platform Enablement	Capability Management; Corporate Support	Platforms, environments, integration, continuity	P10, P06, P02, P05
10	Customer & Stakeholder Engagement	Strategy & Leadership; Corporate Support	Understood needs, service portfolio, relationships, impact	P04, P08, P02

i Note

This consolidated view omits the per-stream “**Applying This to Your NSI**” maturity checklists and scenarios. For those, and for the full Design / Implementation / Management detail, diagrams, and boundary-interface tables, see the individual documents VS-01.0 ... VS-10.0 in this folder.